Guillem Cazassus

Curriculum Vitae

Positions

2025–2027 **Postdoctoral Fellow**, Institute for Mathematical Science, CUHK.

2023–2025 **Postdoctoral Researcher**, Centre for Quantum Mathematics, SDU.

2020–2023 **Postdoctoral Research Associate in Symplectic Topology**, University of Oxford.

2017–2020 **Zorn Postdoctoral Fellow**, Indiana University, Bloomington.

2016–2017 **A.T.E.R**, *(Visiting Assistant Professor)*, Laboratoire de Mathématiques Jean Leray, Nantes.

2015–2016 **A.T.E.R**, *(Visiting Assistant Professor)*, Institut de Mathématiques de Toulouse.

Education

Institut de Mathématiques de Toulouse, Université Paul Sabatier

2012–2016 **Ph.D. Candidate**, *Mathematics*.

- Defense: April 12th, 2016.
- Dissertation Topic: Instanton Symplectic Homology and Dehn Surgery.
- Advisors: Michel Boileau, Paolo Ghiggini.

Ecole Normale Supérieure (ENS) de Lyon

2011–2012 **Agrégation de Mathématiques**, *french teaching examination*.

2009–2011 **M.Sc., Mathematics**, *Master's thesis under the supervision of Michel Boileau*.

2008–2009 **B.Sc., Mathematics**.

Research monographs

In preparation A new construction of Fukaya categories of symplectic manifolds, *G. Cazassus, D. Joyce and A. Ritter*.

Research papers

In preparation Bridgeland stability for Calabi–Yau 2-folds, *G. Cazassus, D. Joyce, J. Lotay and A. Ritter*.

In preparation Donaldson theory, Seiberg-Witten theory and extended field theories, *G. Cazassus, A. Doan and G. Xu*.

In preparation Bialgebras, and Lie monoid actions in Morse and Floer theory, II, *G. Cazassus*.

Preprint The Morse complex is an ∞-functor, arXiv:2505.01362 *G. Cazassus*.

Preprint Bialgebras, and Lie monoid actions in Morse and Floer theory, I, arXiv:2410.16225 *G. Cazassus, A. Hock and T. Mazuir*.

Preprint Equivariant Lagrangian Floer homology via multiplicative flow trees, arXiv:2404.17393 *G. Cazassus*.

Equivariant Lagrangian Floer homology via cotangent bundles of EG_N, arXiv:2202.10097 *G. Cazassus*, **Journal of Topology**.

The correspondence induced on the pillowcase by the earring tangle, arXiv:2010.04320 *G. Cazassus*, *C. Herald*, *P. Kirk*, *A. Kotelskiy*, **Journal of Topology**.

A two-category of Hamiltonian manifolds, and a (1+1+1)-field theory, arXiv:1903.10686 *G. Cazassus*, **Indiana University Mathematics Journal**.

Symplectic Instanton Homology: Naturality, and maps from cobordisms, arXiv:1710.03872 *G. Cazassus*, **Quantum Topology**.

Symplectic Instanton Homology: twisting, connected sums and Dehn surgery, arXiv:1606.00239, *G. Cazassus*, **Journal of Symplectic Geometry**.

Conference proceedings

Tangles, relative character varieties, and holonomy perturbed traceless flat moduli spaces, arXiv:2102.00901 *G. Cazassus*, *C. Herald*, *P. Kirk*, **The Open Book Series**.

Invited talks

2026 IECL Metz, Mathematical Physics Seminar, February 25.

CUHK, Mathematics Inspierd by String Theory Conference, February 12.

2025 UPC Barcelona, Geometry Seminar, March 10.

2024 Krusenberg, Workshop "Uppsala-Odense Geometry Days" , October 24-25.

Rutgers University, Symplectic Geometry Seminar, April 4.

Rutgers University, Gauge Theory Seminar, April 2.

Brussels University, Geometry Seminar, February 26.

Grenoble University, Topology Seminar, February 23.

Strasbourg University, Geometry Seminar, February 12.

2023 SDU Odense, QM Research Seminar, December 4.

Orsay University, Two days of symplectic and contact topology, November 30.

Belalp Hotel, Merging Categorification, Gauge Theory, and Physics conference (short talk), September 8.

Imperial College London, Symplectic Topology Workshop, June 8.

Montpellier University, Algebra, Algebraic Geometry, Algebraic Topology seminar, April 6.

Humboldt University Berlin, Differential Geometry seminar, January 25.

Humboldt University Berlin, Symplectic Geometry seminar, January 23.

2022 UCL London, Oxford-London Gauge Assembly, November 11.

Université de Grenoble, AMS-SMF-EMS International Meeting, July 18-22.

Université libre de Bruxelles, Geometry seminar, April 19.

Institut Henri Poincaré, "Symplectix" seminar, April 1.

2021 James Madison University, colloquium, September 27.

Sofia, 14th ed. Workshop "Lie Theory and Its Applications in Physics", June 20-26.

University of Warsaw, Knot theory seminar, May 14.

Regensburg University, low-dimensional Geometry and Topology seminar, April 27.

Uppsala University, Geometry and Topology seminar, April 7.

Gauge Theory Virtual, March 31.

Gauge Theory, Geometry, and Low-Dimensional Topology special session, AMS Spring Virtual Meeting, March 20-21.

Université Paris-Saclay, Geometry, Topology and Dynamics seminar, March 11.

University of Nevada RENO, Topology seminar, February 5.

University of Oxford, Geometry and analysis seminar, January 25.

2019 Texas AMU, Geometry seminar, November 22.

University of Wisconsin, AMS Sectional meeting, September 15.

University of Illinois, Symplectic Geometry seminar, May 6

IMJ Paris Topology seminar, March 12.

Simons Center for Geometry and Physics, Symplectic Geometry and Toplogy seminar, February 28.

Rutgers University, Geometry and Topology seminar, February 26.

2018 Indiana University Bloomington, Topology and Geometry seminar, September 19.

Indiana University Bloomington, Midwest Topology Seminar, April 28.

Princeton University, Topology seminar, March 15.

Michigan State University, Topology and Geometry seminar, February 1.

2017 MIT, Topology and Geometry seminar, October 23.

Indiana University Bloomington, Topology and Geometry seminar, September 6.

Erlangen University, Algebra and Geometry seminar, July 3.

UCLA, Topology seminar, May 17.

Indiana University Bloomington, Topology and Geometry seminar, May 9.

UQAM (Montreal), Topology and Geometry seminar, May 5.

McMaster University, Topology and Geometry seminar, May 2.

Simons Center for Geometry and Physics, "Gauge Theory and Low Dimensional Topology" Workshop, April 27.

Grenoble University, Topology and Geometry seminar, March 31.

Lille University, "Quantum topology" Workshop, March 9–10.

Nantes University, "Contact and Symplectic Topology" Workshop, January 26–28.

2016 Nantes University, Geometry and Topology seminar, October 10.

Regensburg University, "3-Manifolds and Floer Theories" Workshop, July 19-22.

Toulouse University, Topology and Geometry seminar, January 12.

2015 Institut Henri Poincaré, "Symplectix" seminar, December 4.

Grenoble University, Topology and Geometry seminar, June 19.

Regensburg University, Oberseminar Globale Analysis, April 29.

Geneva University Topology and Geometry seminar, March 12.

2014 Pisa University Topology and Geometry seminar, December 16.

IMJ Paris Topology seminar, October 7.

Nantes University Geometry and Topology seminar, May 5.

Community Service

Seminar and conference organization

2023-2024 With Du Pei, co-organization of the Quantum Lunch reading seminar.

2023 Wrote scientific program for the "Topology between the waves III" winter school on Morse/Floer theory and Fukaya categories.

2022-2023 With Shih-Kai Chiu, Ilyas Khan and Henry Liu, co-organization of Oxford's Geometry and Analysis seminar.

2021-2022 With Shih-Kai Chiu, Ilyas Khan and Henry Liu, co-organization of Oxford's Geometry and Analysis seminar.

2021 With Jason Lotay, co-organization of a reading seminar on Fukaya-Seidel categories and gauge theory.

2020-2021 With Markus Upmeier, co-organization of Oxford's Geometry and Analysis seminar.

2018-2019 With Artem Kotelskiy and Paul Kirk, co-organization of a reading seminar on Fukaya categories.

Outreach

November 2016 Talk at the "5 minutes Lebesgue" on the ham sandwich theorem, video available at: *https://www.youtube.com/watch?v=JqEpURyW_Y4*.

May 2015 "Hippocampe" initiation to research, aimed at high-school students, on "combinatorial games".

Other academic activities

2021 "Mental Health First Aid" training ($\sim$ 20h), University of Oxford.

May 2014 Four lectures on moduli spaces of flat connections and the construction of Symplectic Instanton Homology, for Nantes topology team.

2013–2014 Co-organization of the Toulouse graduate students seminar.

Fall 2013 Nine informal exercise sessions on Algebraic Topology, for Thomas Fiedler's Master's course students.

Fall 2012 Four lectures on Riemannian Geometry and Curvature, for Ph.D. Students in Applied Mathematics, jointly with Jeremy Toulisse.

Teaching Experience

At the Chinese University of Hong Kong:

2025–2026 Student reading seminar supervision, Morse Theory and Fukaya Categories.

At University of Southern Denmark:

2024–2025 Master's course, Riemannian Geometry.

At Indiana University:

2019–2020 M211 (Calculus 1), M722 (Selected topics in Topology: Symplectic Topology and Fukaya categories).

2018–2019 M211 (Calculus 1), M301 (Linear Algebra and Applications).

2017–2018 M211 (Calculus 1), M120 (A brief survey of Calculus 2).

At Nantes University:

2016–2017 First year course and exercise classes for biology undergraduate students. Exercise classes for first year undergraduate students who plan to apply for engineering school.

At Toulouse University:

2015–2016 Oral evaluations for third year math undergraduate students, and exercise classes for first year biology undergraduate students.

2014–2015 First year undergraduate course and exercise classes for students who plan to apply for engineering school. Exercise classes and oral evaluations for Agregation (French teaching examination requiring a Master's degree).

2013-2014 First year undergraduate course and exercise classes. Exercise classes and oral evaluations for Agregation.

2012-2013 Exercise classes for second year undergraduate students who plan to apply for engineering school.

Student supervision

2022-2023 Part B dissertation of Andres Klene-Sanchez, University of Oxford.

Visiting period

Fall 2014 Pisa University (ATUPS grant).

Language Skills

French (Native), English (Fluent), Spanish (Fluent), Catalan (Notions).